A Multi-Fidelity Simulation Framework for X-Band Pulse Compression Radar

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As surface vessels operate in dynamic and cluttered environments, the demand for robust radar-based perception has outpaced the availability of labeled training data. Collecting diverse radar datasets with reliably accurate ground truth is logistically complex, expensive, and constrained by safety considerations, yet modern detection and tracking algorithms require vast amounts of such data to generalize across sea states, object types, and clutter regimes. We present a multi-fidelity simulation framework for X-band radar designed to bridge this gap. The framework comprises three computational tiers: (i) an electromagnetic ray-tracing engine that simulates pulse-level interactions with 3D scene geometry including material-dependent scattering and multipath effects; (ii) an analytical generator using statistical clutter models and stochastic target cross-section representations for rapid, large-scale data generation; and (iii) a data-driven compositor that blends real object signatures into existing radar recordings, preserving real sensor noise and texture. Each tier addresses a different stage of the algorithm development cycle, namely physics-grounded validation, high-throughput training data generation, and domain-transfer verification. We describe the architecture and computational trade-offs of each mode and demonstrate their use for benchmarking detection and tracking methods for small maritime objects across representative sea-clutter and multipath conditions. The framework supports physics-grounded validation, high-throughput synthetic data generation, and domain-transfer checks using recorded radar data, enabling controlled studies of clutter rejection, object identification, and intermittent-return tracking through configurable scene geometry, sensor parameters, and fidelity settings.

I. Introduction

As uncrewed surface vessels (USVs) and autonomous maritime systems increasingly move from experimental prototypes to operational deployment, the demand for robust, all-weather perception capabilities has never been higher [1]. Marine radar, particularly in the X-band (8–12 GHz), remains the primary sensor for long-range obstacle detection and navigation due to its ability to operate effectively in fog, rain, and darkness where optical sensors fail. However, the performance of modern detection and tracking algorithms—especially those driven by deep learning—is heavily dependent on the quantity and diversity of the data used for training and validation [2].

In the maritime domain, data scarcity is a critical bottleneck. Unlike the automotive sector, which benefits from massive open datasets like calibration-quality LiDAR and camera logs (e.g., KITTI, nuScenes), the maritime community lacks comparable resources. Collecting radar data requires specialized hardware, access to vessels, and operation in uncontrolled and often hazardous environments. More importantly, establishing "ground truth" in the open ocean is exceptionally difficult; unlike a car on a road, a vessel at sea has no lane markers, and small targets like kayaks or buoys can be invisible to AIS (Automatic Identification System) and difficult to spot visually or with lidar at radar ranges [3].

Synthetic data generation offers a potential solution, promising unlimited labeled data with perfect ground truth. Yet, simulation in the radar domain is fraught with challenges. Radar returns are not simple geometric projections; they are the result of complex electromagnetic interactions, including multipath propagation, diffraction, and speckle interference. Maritime environments add further complexity with dynamic sea surfaces that generate heavy-tailed, non-stationary clutter that varies with wind speed, direction, and fetch. Simple geometric simulations fail to capture these phenomenologies, leading to a "domain gap" where algorithms trained on synthetic data fail in the real world. Conversely, full-wave electromagnetic solvers are too computationally intensive to generate the thousands of hours of data needed for machine learning.

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To address this dilemma, we propose a scalable, multi-fidelity simulation framework that allows researchers to select the optimal trade-off between physical fidelity and computational cost. Our contribution focuses on three distinct tiers of simulation, each serving a specific phase of the algorithm development lifecycle:

- **Tier 1: High-Fidelity Electromagnetic Ray Tracing.** Theoretically grounded in Maxwell's equations under high-frequency approximations, this tier represents our "Gold Standard." We employ NVIDIA OptiX to accelerate Shooting and Bouncing Rays (SBR), simulating millions of ray interactions per pulse. This tier models complex multipath effects, such as the corner reflector formed between a hull and the sea surface, and incorporates material-specific scattering physics via Physical Optics (PO) and Geometrical Theory of Diffraction (GTD). It is intended for creating validation ground truth and analyzing specific failure modes of detection algorithms.
- **Tier 2: Analytical Approximation.** Designed for speed and massive scale, this tier abstracts the world into statistical distributions. Instead of ray-tracing a complex ocean surface, it synthesizes clutter using the K-distribution or Pareto distribution, parameterized by empirical sea-state models. Targets are injected as point scatterers with RCS fluctuations based on Swerling models. This tier runs orders of magnitude faster than real-time, making it ideal for the inner loops of Reinforcement Learning (RL) training or massive dataset augmentation.
- **Tier 3: Data-Driven Compositor.** Targeting domain adaptation, this tier takes a "cut-and-paste" approach. It extracts real target signatures from field recordings and seamlessly blends them into new background recordings using Poisson image editing. This preserves the exact, messy noise characteristics of the real sensor—thermal noise, quantization artifacts, and receiver saturation—that are notoriously difficult to simulate from first principles. This tier is crucial for verifying that algorithms trained on synthetic data can transfer to the real world.

The remainder of this paper is structured as follows. Section II reviews related work. Section III details the unified system architecture. Sections IV, V, and VI provide rigorous mathematical derivations for the Ray-Tracing, Analytical, and Compositor tiers, respectively. Section VII discusses the specific implementation details, including the GPU kernel architecture. Section VIII presents extensive validation results, Section IX discusses the implications of the "Reality Gap," and Section X concludes.

II. Related Work

A. Electromagnetic Computation in Radar

The gold standard for radar cross-section (RCS) prediction lies in rigorous Computational Electromagnetics (CEM). Full-wave methods like the Method of Moments (MoM) or Finite Difference Time Domain (FDTD) provide exact solutions to Maxwell's equations [4]. However, their computational cost scales with the electrical size of the object to the power of 3 or 4. At X-band frequencies (≈ 9.4 GHz, $\lambda \approx 3$ cm), a simple 10-meter boat spans over 300 wavelengths. Exact solution of such a scene, especially when including a vast sea surface, is computationally intractable for generating dynamic time-series data. High-frequency asymptotic methods offer a practical alternative. Physical Optics (PO), Geometric Optics (GO), and the Uniform Theory of Diffraction (UTD) approximate the scattering behavior when the wavelength is small compared to the target features. The Shooting and Bouncing Rays (SBR) technique [5] combines geometric ray tracing with PO integration on the final bounce, enabling efficient calculation of multi-bounce effects for complex targets. Our work extends SBR from static RCS prediction to full-scene imaging simulation.

B. Atmospheric Propagation Models

Beyond scattering, the propagation of radar waves through the maritime atmosphere is highly non-trivial. Evaporation ducts, formed by the rapid decrease in humidity immediately above the sea surface, create a waveguide effect that can extend radar horizons but also trap clutter energy [11]. Historically, models like the Parabolic Equation (PE) method or the Advanced Propagation Model (APM) have been used to predict path loss in ducting conditions. Most current simulators neglect this, assuming standard $4/3$ earth radius propagation. While our Tier 1 simulator currently uses effective earth radius approximations, the ray-tracing architecture allows for future integration of refractive index gradients $\eta(z)$ by curving rays in discrete steps, a method known as "ray marching" in computer graphics.

C. Maritime Clutter Modeling

Modeling sea clutter—the unwanted backscatter from the ocean surface—is a field with a rich history. While low-resolution radars may observe Rayleigh-distributed clutter, high-resolution systems operating at low grazing angles observe deviation from Gaussian behavior, characterized by heavy tails or "spikiness." The K-distribution has become the standard model for such clutter, treating the return as a product of two independent random variables: a fast-fluctuating Rayleigh component (speckle) and a slowly varying Gamma component (texture) [7]. More recent models like the Pareto and KA-distribution offer better fits for very spiky clutter [8]. Our Tier 2 simulator directly implements these statistical models, generating spatially correlated realizations to mimic the "patchiness" of ocean swells.

D. Deep Learning in Radar

The application of Convolutional Neural Networks (CNNs) to radar imagery has grown exponentially. Works like [10] have demonstrated the efficacy of CNNs for target classification, but note the severe performance drop when training data is limited. Generative Adversarial Networks (GANs) have been proposed to augment radar datasets, but ensuring physical consistency—such as the correlation between a target's orientation and its RCS—is difficult for black-box generative models. Our Tier 3 approach avoids this "hallucination problem" by using real physics-derived data snippets.

E. Game Engines for Simulation

The autonomous driving community has heavily leveraged game engines like Unreal Engine and Unity to simulate camera and LiDAR data (e.g., CARLA, AirSim) [9]. While these engines provide photorealistic visual rendering, their built-in physics are typically limited to simple ray casting for collision detection. They typically do not model phase accumulation, polarization, or diffraction, which are fundamental to radar. While recent plugins have attempted to add radar capabilities to these engines, they often rely on simple geometric buffers (G-buffers) and screen-space shaders, failing to capture true multi-bounce multipath. Our approach bypasses standard game engine rendering in favor of a custom OptiX-based pipeline that prioritizes electromagnetic validity over visual appearance.

III. System Architecture and World Model

The framework is built on a "Model Once, Simulate Anywhere" philosophy. A single scenario definition feeds all three simulation tiers, ensuring geometric and semantic consistency.

A. Unified Scene Graph

The world state is managed by a central scene graph containing:

- **Ego Vector**: The kinematic state of the sensor platform $\mathbf{x}_s(t) = [x, y, z, \phi, \theta, \psi, \dot{x}, \dot{y}, \dot{z}, \dot{\phi}, \dot{\theta}, \dot{\psi}]^T$.
- **Static Geometry**: Terrain meshes derived from high-resolution DEM (Digital Elevation Models) and detailed CAD models for static infrastructure (buoys, markers, pilings).
- **Dynamic Entities**: Moving vessels, each characterized by a 3D mesh, a rigid-body trajectory, and material properties (conductivity, permittivity).
- **Environmental State**: A description of the sea state (significant wave height $H_{1/3}$, dominant period T_p, wave direction Ω_w), wind vector, and time of day.
- **Sensor Parameters**: Transmit frequency f_c, pulse width τ_p, antenna beam pattern $G(\theta, \phi)$, transmit power P_{tx}, and system losses L_{sys}.

B. Output Standardization

To ensure seamless integration with downstream processing stacks, all tiers output data in a standardized format mimicking the Furuno NXT solid-state radar digital video stream. This sensor utilizes Pulse Compression (PC) to achieve high range resolution with low peak transmit power.

- **Format**: Polar coordinate stream.
- **Resolution**: 4096 azimuth spokes per rotation (0.088° resolution) × 2048 linear range bins.
- **Quantization**: 8-bit unsigned integer (0-255) linear intensity.

This standardization allows direct "drop-in" replacement of real radar drivers with our simulator drivers for Hardware-in-the-Loop (HIL) testing.

IV. Tier 1: High-Fidelity Electromagnetic Ray Tracing

The Tier 1 simulator is a physics-based engine designed to capture the detailed propagation of electromagnetic waves. It solves the radar range equation in its integral form via Monte Carlo sampling.

A. Ray Generation and Importance Sampling

We model the antenna as a point source located at $\mathbf{r}_{tx}$. We launch $N_{rays} \approx 10^6$ rays per pulse. To maximize efficiency, we employ importance sampling. Rays are not distributed uniformly on the sphere; rather, their launch directions (θ, ϕ) are sampled from a probability density function proportional to the antenna's gain pattern $G(\theta, \phi)$. This ensures that computing resources are focused on the main lobe and significant sidelobes, rather than wasted on the backlobes where energy is negligible. For a given ray i, the initial electric field amplitude is:

$$E_0^i = \sqrt{\frac{P_{tx} G(\theta_i, \phi_i) Z_0}{4\pi}} \tag{1}$$

where $Z_0 \approx 377\Omega$ is the impedance of free space.

B. Dynamic Sea Surface Synthesis

A static mesh is insufficient for maritime radar. We synthesize a dynamic ocean surface $\eta(x, y, t)$ using a spectral method. We create a wave spectrum $S(\mathbf{k})$ (e.g., JONSWAP or Pierson-Moskowitz) representing the distribution of wave energy variance over wave numbers $\mathbf{k}$.

$$S(\omega) = \frac{\alpha g^2}{\omega^5} \exp\left[-\frac{5}{4}\left(\frac{\omega_p}{\omega}\right)^4\right] \gamma^{\exp\left[-\frac{(\omega-\omega_p)^2}{2\sigma^2\omega_p^2}\right]} \tag{2}$$

The surface elevation is obtained via IFFT of complex amplitudes modulated by a random phase:

$$\eta(\mathbf{x}, t) = \mathrm{Re}\left[\sum_{\mathbf{k}} \sqrt{S(\mathbf{k})\Delta\mathbf{k}}\, e^{i(\mathbf{k}\cdot\mathbf{x} - \omega(k)t + \Phi)}\right] \tag{3}$$

For Tier 1, this height field is tessellated into a triangle mesh on the GPU at every simulation time step. This allows us to capture the time-varying shadowing and tilt modulation of sea clutter.

C. Electromagnetic Interaction Models

We implement a hybrid PO/GTD/GO scattering model.

1. Physical Optics (PO)

For electrically large, smooth surfaces (like hull plates), PO approximates the induced surface current $\mathbf{J}_s$ on the lit region as $\mathbf{J}_s \approx 2\hat{n} \times \mathbf{H}_{inc}$, where $\mathbf{H}_{inc}$ is the incident magnetic field. The scattered electric field $\mathbf{E}_s$ in the far field is:

$$\mathbf{E}_s(\mathbf{r}) = -jkZ_0 \frac{e^{-jkr}}{4\pi r} \iint_S \hat{r} \times (\hat{r} \times \mathbf{J}_s) e^{jk\hat{r}\cdot\mathbf{r}'} dS' \tag{4}$$

In the ray-tracing context, this integral is approximated by the sum of contributions from each ray "hit," effectively performing a Monte Carlo integration over the illuminated surface area.

2. Geometrical Theory of Diffraction (GTD)

PO predicts zero scattering in the geometric shadow and fails to accurately model sharp wedges. We augment PO with GTD to handle edge diffraction. When a ray strikes near an edge, we calculate a diffraction coefficient $\mathbf{D}(L, \beta_0, n)$ based on the wedge angle and incidence direction.

$$\mathbf{E}_d = \mathbf{E}_{inc} \cdot \mathbf{D} \cdot \sqrt{\frac{1}{r(1 + r/\rho)}} e^{-jkr} \tag{5}$$

This allows us to simulate returns from masts, antennas, and railings, which are often the dominant scatterers on stealthy vessels.

D. Multi-Bounce and Corner Reflectors

One of the most critical phenomena in maritime radar is the dihedral corner reflector effect formed by the intersection of a vertical ship hull and the horizontal sea surface. This structure yields a massive backscatter return, often orders of magnitude larger than the direct return from the hull alone. Tier 1 captures this by tracing rays up to a depth of $N_{bounce} = 4$. 1. Ray hits water surface → reflects towards ship. 2. Ray hits ship hull → reflects back towards radar. This "double-bounce" path length is coherently added to the "direct" path length, resulting not just in increased power but also in critical phase interference effects.

E. Coherent Summation and Speckle

Radar is a coherent sensor. The voltage V measured at the receiver for a specific range bin is the complex vector sum of all field contributions E_m falling into that bin:

$$V = \left| \sum_{m=1}^{M} |E_m| e^{j\phi_m} \right| \tag{6}$$

where the phase $\phi_m = -2kR_m + \Delta\phi_{scat}$. Because k is large, tiny variations in range R_m (on the order of millimeters) cause large phase rotations. This naturally reproduces *speckle*—the granular, noisy texture of coherent imagery—without any artificial noise injection. It correctly models the fading fluctuations seen as a target rotates or as waves move.

F. Pulse Compression Simulation

The Furuno NXT is a solid-state radar employing Pulse Compression (PC). Instead of a short, high-power pulse, it transmits a long Frequency Modulated (FM) chirp to maximize total energy on target while maintaining range resolution. In Tier 1, we simulate this by transmitting a ray bundle corresponding to the long pulse duration τ_{raw}. The received field $E_{rx}(t)$ is then convolved with the matched filter $h(t) = s^*(-t)$ of the transmitted chirp $s(t)$.

$$V_{out}(t) = E_{rx}(t) * h(t) \tag{7}$$

This operation compresses the energy into a narrow main lobe of width $\tau_{comp} \approx 1/B$, where B is the chirp bandwidth. This allows us to capture range-sidelobe artifacts which are characteristic of solid-state radars and often mistaken for real targets in high-dynamic-range scenarios.

V. Tier 2: Analytical Approximation

While Tier 1 is physically rigorous, it is computationally heavy. Tier 2 is designed for high-throughput training, abstracting physics into statistics.

A. Statistical Clutter Synthesis

We synthesize the sea clutter background directly in the image domain. We employ the Compound K-distribution model, widely validated for high-resolution maritime radar. The clutter intensity I is modeled as:

$$I(n) = \sqrt{\tau(n)} \cdot u(n) \tag{8}$$

where $u(n)$ is a complex Gaussian process representing speckle (due to many small capillary waves) and $\tau(n)$ is a Gamma-distributed texture process (due to large-scale gravity waves tilting the surface).

$$p(\tau) = \frac{1}{\Gamma(\nu)} \left(\frac{\nu}{\mu}\right)^{\nu} \tau^{\nu-1} e^{-\frac{\nu\tau}{\mu}} \tag{9}$$

Here, μ is the mean power and ν is the shape parameter. Low ν $(0.1 - 1.0)$ corresponds to "spiky" clutter (high sea states), while high ν (> 10) approaches Gaussian noise (calm seas).

Crucially, the texture τ is not independent pixel-to-pixel; it exhibits spatial correlation corresponding to the wave structure. We generate correlated texture by convolving white Gamma noise with a kernel derived from the sea wave spectrum, or by using Spherically Invariant Random Processes (SIRP). The result is a field of clutter that mimics the visual structure of ocean swells, key for training Deep Learning algorithms (e.g. CNNs) that rely on texture analysis.

B. Target Injection Model

Targets are inserted as additive signal sources. We model the target RCS $\sigma(t)$ as a stochastic process:

$$\sigma(t) = \sigma_{avg} \cdot f_{aspect}(\theta_{obs}) \cdot S(t) \tag{10}$$

- σ_{avg}: The nominal average RCS of the vessel.
- f_{aspect}: A lookup table or function modulating RCS based on aspect angle (e.g., broadside flash).
- $S(t)$: A fluctuation term drawn from Swerling chi-squared distributions. Swerling 1/3 (slow fluctuation) are drawn once per scan; Swerling 2/4 (fast fluctuation) are drawn per pulse.

The propagation loss is computed via the standard two-way radar equation, including atmospheric attenuation α_{atm} (dB/km).

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4} 10^{-0.2\alpha_{atm} R} \tag{11}$$

This received power is converted to a digital number (DN) and added non-coherently to the background clutter power. This simplifies the interference but allows for extremely fast computation.

VI. Tier 3: Data-Driven Compositor

Tier 3 addresses the "sim-to-real" gap by using real data as the substrate.

A. Background and Target Libraries

We maintain a library of "Backgrounds"—long recordings of open ocean or harbors scrubbed of targets. We also maintain a "Target Chip Application" (TCA) database. These chips are extracted from real data: 1. Detect high-SNR target in real log. 2. Extract $N \times M$ window around target. 3. Segment target from clutter using alpha-matting or constant-false-alarm thresholding. 4. Normalize intensity by estimated range.

B. Intensity-Aware Blending

To insert a target chip into a new background, simple addition is insufficient. We perform: 1. **Range Scaling**: The chip intensity is scaled by $(R_{src}/R_{dest})^4$ to account for geometric spreading. 2. **Resampling**: The polar chip is resampled to match the angular/range resolution at the new range. 3. **Poisson Blending**: We solve the Poisson equation $\Delta I = \mathrm{div}(\mathbf{v})$ over the target domain Ω, where $\mathbf{v}$ is the gradient field of the source target. Boundary conditions are set by the destination background pixels. This seamlessly integrates the target into the clutter, matching local contrast and eliminating "paste lines."

C. Shadow Implantation

A major limitation of naive pasting is the lack of proper shadowing. A target placed in front of a heavy clutter patch should occlude it. We approximate this effect by defining a "shadow sector" behind the target in polar space. We apply a multiplicative gain mask $M(r, \theta) < 1.0$ to the background pixels in this sector.

$$M(r, \theta) = 1 - \alpha_{shadow} \cdot \exp\left(-\frac{|\theta - \theta_{target}|}{\sigma_\theta}\right) \tag{12}$$

This creates the characteristic dark streak behind large vessels, crucial for training algorithms to recognize occlusion.

VII. Implementation Details

A. GPU Ray Tracing Kernel (Tier 1)

The Tier 1 engine is built on **NVIDIA OptiX 7.0** for hardware-accelerated intersection. The pipeline consists of three CUDA programs:

- **RayGen**: Samples launch directions and initializes ray payloads (field amplitude, phase, depth).
- **ClosestHit**: Invoked upon surface intersection. It evaluates the material BRDF, calculates the path phase accumulated, spawns reflection rays, and uses atomic operations to accumulate the field into the receiver buffer.

- **Miss**: Handles rays that escape to the sky (no return).

To handle the dynamic sea surface, we invoke a CUDA kernel at the start of each frame that computes the FFT of the wave spectrum and updates the vertices of a pre-allocated triangle grid. A OptiX Acceleration Structure (AS) rebuild is triggered—modern GPUs can rebuild a 10^6 triangle BVH in milliseconds.

The algorithm for the RayGen program is outlined below:

Algorithm 1 Ray Generation Kernel (Simplified)

function RayGen
  $idx \leftarrow$ optixGetLaunchIndex
  $(\theta, \phi) \leftarrow$ SampleAntennaPattern(idx)
  $\mathbf{d} \leftarrow$ SphericalToCartesian(θ, ϕ)
  $\mathbf{origin} \leftarrow \mathbf{Pos}_{radar}$
  $E_0 \leftarrow$ ComputeInitialField(θ, ϕ)
  $payload \leftarrow \{E_0, \text{phase} = 0, \text{depth} = 0, \text{path\_len} = 0\}$
  optixTrace$(\mathbf{origin}, \mathbf{d}, payload)$
  $I \leftarrow |payload.finalE|^2$
  AtomicAddOutput(I)
end function

B. Analytical Engine (Tier 2)

Tier 2 is implemented in **Rust** using the `Rayon` library for multithreading. Since each azimuth spoke is independent (assuming constant sea state), the problem is "embarrassingly parallel." We use SIMD intrinsics to accelerate the Gaussian number generation and Gamma variate synthesis. The statistical synthesis of a full 4096-spoke frame takes < 50 ms on a modern CPU, enabling massive parallel data generation.

C. Software Architecture

The framework exposes a unified Python API wrapping the underlying Rust/C++ engines. This integration allows users to seamlessly switch backends:

```
sim = RadarSim(config="scenario_1.yaml")
# Switch backend at runtime
sim.set_tier(Tier.RAY_TRACING)
frame_high_fi = sim.step()

sim.set_tier(Tier.ANALYTICAL)
frame_fast = sim.step()
```

VIII. Results and Validation

A. Simulation Fidelity Analysis

We validated the clutter fidelity by comparing the amplitude statistics of simulated frames against real X-band data collected from a shore station in Charleston, SC. We extracted clutter patches and fit the K-distribution shape parameter ν. The results (Table 1) show that the Ray Tracer organically produces non-Gaussian clutter ($\nu < 10$) as sea roughness increases, purely through geometric interaction with the wave surface, validating the physical basis of the model.

B. Computational Performance

We benchmarked the three tiers on an NVIDIA RTX 4090. Tier 1 is clearly offline-only, but its physical accuracy is unmatched. Tiers 2 and 3 enable real-time and even faster-than-real-time simulation, critical for training agents that require millions of experience steps.

Table 1 Clutter Statistics Comparison (Shape Parameter ν)

Sea State	Real Data	Tier 1 (Ray)	Tier 2 (Analytical)
Calm (SS 1)	10.5	12.1	10.0 (Fixed)
Choppy (SS 3)	2.1	2.4	2.0 (Fixed)
Rough (SS 5)	0.8	0.85	0.8 (Fixed)

Table 2 Runtime Performance Metrics

Tier	Latency	Throughput	Use Case
1 (Ray)	45.0 s/sweep	0.02 Hz	Validation / Ground Truth
2 (Analytic)	0.05 s/sweep	20 Hz	ML Training / RL
3 (Compos)	0.005 s/sweep	200 Hz	Domain Transfer Check

C. Detection Algorithm Benchmarking

We ran a standard CFAR (Cell-Averaging Constant False Alarm Rate) detector on 1000 frames from each tier and real data to compare detection probability (P_d) for a benchmark target (0.5 m^2 RCS buoy).

- **Real Data**: $P_d = 0.78$ at $P_{fa} = 10^{-4}$.
- **Tier 1**: $P_d = 0.81$. The slight overestimation is likely due to perfect material assumptions.
- **Tier 2**: $P_d = 0.85$. The analytical model tends to be "cleaner" than reality, making targets slightly easier to distinguish.
- **Tier 3**: $P_d = 0.77$. The compositor achieved the closest match, likely because the background noise floor was real.

D. Visual Comparison

(a) Tier 1: High Fidelity **(b) Tier 2: Analytical** **(c) Real Data Reference**

Fig. 1 Visual comparison of simulated PPI scopes vs real data. Tier 1 captures realistic shadowing and edge diffraction. Tier 2 provides a cleaner statistical realization.

IX. Case Study: Charleston Harbor Simulator

To demonstrate the utility of the framework, we constructed a digital twin of the Charleston Harbor entrance.

A. Environment Construction

We imported high-resolution bathymetry data from NOAA (National Oceanic and Atmospheric Administration) and LiDAR scans of the jetties. The simulation region covers a 10 km × 10 km area. Static objects such as channel markers were placed at their precise GPS coordinates.

B. Scenario Definition

We simulated a "Collision Avoidance" scenario where the ego-vessel (a 7m USV) encounters a high-speed intruder (a jet ski) obscured by heavy clutter near the jetty rocks. Using Tier 1, we were able to visualize the specific multipath interference caused by the jetty rocks, which often creates "ghost targets" in real radar data. This phenomenon is critical for testing tracking algorithms that rely on kinematic consistency to filter out false alarms. Using Tier 2, we generated 10,000 variations of this encounter with randomized intruder speeds and headings, providing a robust training set for an RL-based collision avoidance agent.

X. Discussion: The Model-Reality Gap

A key finding of this work is the trade-off between statistical precision and structural accuracy.

A. The Texture vs. Structure Trade-off

Tier 2 (Analytical) yields perfect statistical fits (by definition) but lacks structural coherence. For example, it cannot simulate the deterministic way a large wave might simultaneously occult a target and create a strong clutter return in front of it. Tier 1 (Ray Tracing) captures this structural interaction but often produces "cleaner" textures than reality due to the lack of un-modeled noise sources (thermal noise, quantization jitter, wind shear).

B. Implications for Deep Learning

Algorithms trained purely on Tier 2 data tended to overfit to the noise distribution. When tested on real data, they failed when clutter distributions shifted slightly (e.g., from K-distribution to Pareto). Algorithms trained on Tier 3 (Composited) data showed much higher robustness, suggesting that "authentic noise" is a critical feature for neural network generalization.

C. Hybrid Training Strategies

We propose a "Curriculum Learning" approach: 1. **Phase 1**: Train agent on 10^7 frames of Tier 2 data to learn basic kinematic tracking. 2. **Phase 2**: Fins-tune on 10^5 frames of Tier 3 data to adapt to real sensor noise profiles. 3. **Phase 3**: Validate on Tier 1 corner-cases to ensure safety constraints are met.

XI. Conclusion and Future Work

We have presented a scalable, multi-fidelity simulation framework for maritime radar that addresses the critical data bottleneck in developing autonomous systems. By offering three distinct tiers, we allow developers to choose the right tool for the task: Ray tracing for rigorous physics validation, Analytical models for massive-scale reinforcement learning, and Compositing for final domain-adaptation checks.

Future work will focus on:

- **Atmospheric Ducting**: Incorporating evaporation duct models (PJ/NPS models) into the ray tracer is essential for modeling Over-The-Horizon (OTH) detection. In maritime environments, the trapping of radar energy in a surface duct can extend detection ranges significantly but also introduces "sea clutter rings" and range-ambiguous returns.
- **Doppler Simulation**: Extending the engine to support coherent radar simulation. By tracking the exact phase rate of change of every ray, we can simulate the micro-Doppler signatures of rotating radar antennas and target vibrations, enabling research into classification of targets by their vibration spectra.
- **Generative AI**: Investigating the use of Diffusion Models to replace the Tier 3 compositor for even more realistic target in-painting.

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